

When can the swarm tell who helped?

Enrico Yao-Bate

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1 Question

A learner receives m messages in one generation. The organization wants, for each message j , its causal effect on the learner, because that number is what pays teachers and decides who talks to whom. With $m = 3$ no cheap estimator recovered the planted effects and only full counterfactual replay did (`findings/2026-10-07-credit-estimators.md`). The question here is *why*, and therefore at what m , and under what kind of messages, credit from one or two replays is recoverable. The answer decides whether “credit agents for their students’ improvement” is a mechanism we can build or a slogan.

2 Setting

One student with a weak incumbent (self-play ≈ 3), verification on (self-play on 200 deals, adopt if better), the local model at temperature 0 with a fixed sampling seed, so that the student’s update is a deterministic function of which messages were delivered. Reaction rule: read everything (the baseline).

The m messages are the 17-point reference bot with plain prose (the good message), plus $m - 1$ others of one of two kinds, which is the factor that matters:

- **neutral**: the student’s own bot re-sent with a neutral note (“here is my current strategy”);
- **one persuasive bad**: the 0-point bot with persuasive prose and honest low evidence, and the rest neutral.

3 What we measure

Write $Y(S)$ for the student’s held-out paired improvement when exactly the subset $S \subseteq [m]$ is delivered, on evaluation deals the model never sees. The truth for message j is its average effect over the delivery of the others,

$$\tau_j = \frac{1}{2^{m-1}} \sum_{S \subseteq [m] \setminus \{j\}} (Y(S \cup \{j\}) - Y(S)),$$

observable exactly by replaying all 2^m subsets under the same seed. The cheap estimator is one marginal, leave-one-out, $\hat{\tau}_j^{\text{LOO}} = Y([m]) - Y([m] \setminus \{j\})$, which costs $m + 1$ replays, and its mirror, leave-one-in, $Y(\{j\}) - Y(\emptyset)$. Both equal τ_j when the messages do not interact; otherwise each is one of the 2^{m-1} marginals that τ_j averages, and its error is the interaction it does not see. That is the whole mechanism, and it is what the factor above tests: a neutral message should not interact with

the good one; a persuasive bad message did (its interaction with the good message was +3.8 points with $m = 3$). Per seed we report RMSE over the m messages for each estimator, and the ratio of the mean $|\tau_j|$ to the between-seed standard deviation of τ_j , which is what any pooled estimator is up against.

4 Prediction (2026-10-08)

1. At $m = 1$ one replay is the truth: leave-one-out RMSE is evaluation noise only (≈ 0.3 , from 300 paired deals with per-deal SD about 4).
2. In the neutral arm, leave-one-out stays within 1 point of the truth at $m = 2$ and $m = 3$: re-sent own bots barely interact with the good message (the re-sent bot’s effect was $+0.7 [-0.5, 1.6]$ and its interaction small).
3. In the persuasive arm, leave-one-out fails already at $m = 2$: its error on the good message is half the good–bad interaction, about 1.9 points, and it does not improve with m . The cause of the earlier failure was not three messages; it was one persuasive message.
4. The between-seed spread of τ for the good message stays near 2 points in both arms, so pooling across seeds cannot rescue per-seed credit in either.

What would make us drop the idea: leave-one-out error above 1 point in the neutral arm at $m = 2$. Then messages interact even when they say nothing, credit needs full replay at any $m > 1$, and the organization must be designed around one message per learner per generation.

5 Design

Factors: $m \in \{1, 2, 3\} \times \text{arm} \in \{\text{neutral}, \text{persuasive}\}$, full factorial, paired on seeds (the same 10 experiment seeds in every cell; a seed fixes the feedback traces the model sees and the evaluation deals). Truth by exhaustive replay: $2 + 4 + 8 = 14$ replays per seed per arm, 280 local calls in all, one night, \$0. The null is the stub in null mode (revisions keep the rule list, no text effect), where leave-one-out is exact; it gives the band that evaluation noise alone produces. Figure: RMSE of leave-one-out and leave-one-in against m , one line per arm, seed points, IQM with a bootstrap interval, the null band shaded. Decision rule, fixed now: prediction 2 holds if the neutral-arm IQM of leave-one-out RMSE at $m = 3$ is ≤ 1 with interval upper bound ≤ 1.5 ; prediction 3 holds if the persuasive-arm IQM at $m = 2$ is ≥ 1.5 with lower bound ≥ 1 . A follow-up at $m = 5$ (singles-and-pairs replay as the truth proxy, 16 replays) runs on a hosted model if both hold.

6 Cost

280 local calls at about 70 s each: 5.5 hours, serial, \$0. Engine time negligible. Hosted follow-up at $m = 5$: $10 \text{ seeds} \times 2 \text{ arms} \times 16 \text{ replays} = 320 \text{ calls}$, a few dollars.

7 Nearest prior work

Counterfactual-replay credit for multi-agent language-model systems masks one agent and replays (SHARP, arXiv 2602.08335); singles-and-pairs search for failure localization (Li et al. 2026, arXiv 2608.29228); Shapley credit with process rewards (Yang et al. 2025, arXiv 2511.10687). None asks

when a one-replay estimator is adequate as a function of how many messages a learner sees and whether they interact; not found in the 2026-10-06 search.